Sophie Bini

Postdoctoral Researcher · California Institute of Technology

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Work experience

California Institute of Technology — *Pasadena, USA*

Postdoctoral researcher — *2024 - present*

- Data analysis for the LIGO Laboratory

Education

University of Trento — *Trento, Italy*

PhD Physics — *2020 - 2024*

- Advisor: Prof. G. A. Prodi
- Thesis: 'Enhancing the detection and the reconstruction of gravitational-wave transients in the LIGO-Virgo-KAGRA data using weak assumptions on the astrophysical sources'
- PhD representative: member of the Department's Council, organizer of PhD workshops

University of Pisa — *Pisa, Italy*

M.Sc. Physics — *2018 - 2020*

- Advisors: Prof. F. Fidecaro, Prof. M. Razzano
- Thesis: 'Unsupervised classification of short transient noise to improve gravitational wave detection'
- Grade: 110/110 cum laude

University of Trento — *Trento, Italy*

B.Sc. Physics — *2014 - 2017*

- Advisor: Prof. G. A. Prodi
- Thesis: 'First detection of electromagnetic counterpart in a gravitational wave event'

Visiting

April - May 2023	**Visiting Program**, University of Zurich, Pauli Center (CH). Advisor: Dr. S. Tiwari
Feb - June 2020	**Erasmus program**, University of Glasgow (UK) for master thesis research program. Advisor: Dr. Messenger.

Grants

2025	**Grant proposal contributor**, NSF Award number 2608024 'Gravitational wave physics and astrophysics with LIGO', PI: Prof. Alan J. Weinstein. Significant written contribution to the awarded grant (roughly 1/3 of the research proposal content)	*$ 480,000.00*
2024	**Grant proposal contributor**, NSF Award number 2513294 'Gravitational wave physics and astrophysics with LIGO', PI: Prof. Alan J. Weinstein. Significant written contribution to the awarded grant (roughly 1/3 of the research proposal content)	*$ 200,000.00*

Awards, Hackathons

2025 **Prize in memory of Stefania Marassi**, prize for early-career female researchers in the field of astrophysics, Sapienza Universita' di Roma

2021 **Premio Tito Maiani**, annual prize for M.Sc. Thesis on fundamental interactions, Accademia Nazionale dei Lincei

2021 **Winner team of 'Industrial Problem Solving with Physics'**, annual hackathon, University of Trento and TechnoAlpin company

2019 **Winner team of the 'NASA space apps challenge - Pisa'**, University of Pisa, https://2019.spaceappschallenge.org/challenges/earths-oceans/trash-cleanup/teams/albatross/project

Research activities

- Member of the LIGO-Virgo-KAGRA Collaboration since 2020.
- Detection and interpretation of high-mass binary black hole mergers, in particular major contribution in the analysis of the gravitational-wave event GW231123.
- Detection and reconstruction of gravitational-wave burst with minimal assumptions. Development of the algorithm CoherentWaveBurst.
- Mitigation of transient noise (glitches) overlapping with gravitational-wave signals.
- Virgo early-career scientist (VECS) representative at the Virgo Steering Committee (VSC) (2024).

Conferences

Invited seminars

- Listening to the Cosmos: New Frontiers in Gravitational Wave Physics, Galileo Galilei Institute for Theoretical Physics, September 2026, 'Current challenges in glitch mitigation for robust black hole spin measurements'
- Multiple Systems Across Scales, Heidelberg University, April 2026, 'Observing intermediate-mass black holes with gravitational waves: recent results using LIGO-Virgo-KAGRA observations and challenges for the future '
- Sapienza Universita' di Roma, January 2026, 'GW231123: a binary black hole with total mass $\sim 190 - 265 M_\odot$'
- GWPAW meeting, Atlanta, December 2025, 'GW231123: a binary black hole with total mass $\sim 190 - 265 M_\odot$'
- KAGRA meeting, Toyama (online), August 2025, 'GW231123: a binary black hole with total mass $\sim 190 - 265 M_\odot$'
- Amaldi/GR meeting, Glasgow, July 2025, 'New results from the LIGO, Virgo and KAGRA Observatory Network'
- UIB University of the Balearic Islands (online), April 2024, 'Search for hyperbolic encounters of compact objects in the third LIGO-Virgo-KAGRA observation run'
- California Institute of Technology (online), February 2024, 'Recent progresses in the detection and reconstruction of GW transients with minimal assumptions'
- UMAss Dartmouth University (online), September 2023, 'Search for hyperbolic encounters of compact objects in the third LIGO-Virgo-KAGRA observation run'

Talks

- 42nd Jim Isenberg Pacific Coast Gravity Meeting, Riverside, March 2026, 'The impact of waveform systematics and Gaussian noise on the interpretation of GW231123'
- LIGO-Virgo-KAGRA meeting, Melbourne, March 2025, 'Update on the exceptional event GW231123' and 'All-sky search for short gravitational-wave bursts in the first eight months of the fourth Advanced LIGO-Virgo-KAGRA observing run'
- APS April Meeting, Anaheim, March 2025, 'Search for hyperbolic encounters of compact objects in the third LIGO-Virgo-KAGRA observation run'
- EPS High Energy Physics conference, Hamburg, August 2023, 'Search for hyperbolic encounters of compact objects in the third LIGO-Virgo-KAGRA observation run'

- Amaldi Conference, online, July 2023, ‘Search for hyperbolic encounters of compact objects in the third LIGO-Virgo-KAGRA observation run’
- Virgo week, Pisa, April 2023, ‘Taking advantage of Virgo in short-duration bursts searches with Coherent WaveBurst’
- LIGO-Virgo-KAGRA meeting, NorthWestern University (online), March 2023, ‘An autoencoder neural network integrated into gravitational-wave burst searches to improve the rejection of noise transients’
- GRASS Conference, Padova, June 2022, ‘Improving future searches for gravitational wave transients on multi-detector data’
- PHAROS 2022 Conference, Rome, May 2022, ‘Search for gravitational wave transients associated with magnetar bursts during the third Advanced LIGO and Advanced Virgo observing run’
- Virgo Week, online, November 2021, ‘Enhancing short duration all-sky burst search with machine learning’
- TAUP Conference, (online), September 2021, ‘Preparing for the future gravitational wave burst searches with machine learning techniques’

Posters

- LIGO-Virgo KAGRA meeting, Toyama, September 2023, ’Search for hyperbolic encouters of compact objects in the third LIGO-Virgo-KAGRA observation run’
- Recontres de Moriond, La Thuile, March 2023, ‘Search for gravitational wave transients associated with magnetar bursts during the third Advanced LIGO and Advanced Virgo observing run’

Teaching

2025	**Workshop**, Transform your research into teaching, one-week long workshop, California Institute of Technology
2021, 2022	**Electronics laboratory course**, (B.Sc. Physics) Teaching Assistant, Universita’ di Trento
2021	**Optics laboratory course**, (B.Sc. Physics) Teaching Assistant, Universita’ di Trento
2016	**Experimental physics course**, (B. Eng.) Teaching Assistant, Universita’ di Trento

Mentoring

2026	**Samantha Walker**, LIGO SURF student (10-weeks summer program for undergraduate students)
2025	**Krzysztof Król**, LIGO SURF student

Outreach

2026	**Watson lecture pre-show**, organization of outreach activities for the pre-show of Katie Bouman’s lecture	*Pasadena*
2026	**Caltech Postdoc Launch Seminars**, talk: ’Gravitational-wave astronomy’	*Pasadena*
2026	**Jet Propulsion Laboratory Camp Day at Dodger stadium**, LIGO demonstration	*Los Angeles*
2025	**Explore Caltech**, public talk: ‘What are black holes?’	*Pasadena*
2025	**Watson lecture pre-show**, organization of outreach activities for the pre-show of Katerina Chatziioannou’s lecture	*Pasadena*
2022	**Participation to the festival of theatre and science ‘Theatre of Wonder’**, writer and performer, in collaboration with Caltech Theater	*Trento*
2020-2026	**Talks in schools**, Scuola secondaria di primo grado ”S.Bellesini” (Vezzano), Scuola Internazionale Bilingue Maria Montessori (Mantova), Girls Academic Leadership Academy (Los Angeles), Blair Middle-high school, Paradise Canyon Elementary Science school (Pasadena), La Cañada High School (Los Angeles).	*Trento, Mantova, Pasadena*

Short author-list publications

- Pearson N., Çalışkan M., **Bini S.**, Cornish N., (2026). A compact time-frequency representation for gravitational-wave data analysis. arXiv preprint 2609.05601
- **Bini S.**, Król K., Chatziioannou K., Isi M., (2026). The impact of waveform systematics and Gaussian noise on the interpretation of GW231123. Physical Review D, 113, 083036
- Gupte N. , Coleman Miller M. , Udall R., **Bini S.**, ... (2026). Eccentricity constraints disfavor single-single capture in nuclear star clusters as the origin of all LIGO-Virgo-KAGRA binary black holes. Accepted in ApJ, arXiv preprint 2603.29019
- Udall R. , **Bini S.**, Chatziioannou K., Miller S., ..., Davis D., (2025). Inferring the spins of merging black holes in the presence of data-quality issues. Physical Review D, 113, 042005
- Martini A., Miani A., Drago M., Lazzaro C., Salemi F., **Bini S.**,..., (2025). Optimizing searches for gravitational wave bursts using coherent WaveBurst 2G. Classical and Quantum Gravity, 43, 055016
- **Bini S.**, Tiwari S., Xu Y., Smith L., Ebersold M., ..., Prodi G.A., (2024). Search for hyperbolic encounters of compact objects in the third LIGO-Virgo-KAGRA observation run. Physical Review D, 109, 042009.
- **Bini S.**, Vedovato G., Drago M., Salemi F., Prodi G.A., (2023). An autoencoder neural network integrated into gravitational-wave burst searches to improve the rejection of noise transients. Classical and Quantum Gravity, 40, 135008
- Szczepańczyk M. J., Salemi F., **Bini S.**, Mishra T., ... Klimenko S. (2023). Search for gravitational-wave bursts in the third Advanced LIGO-Virgo run with coherent WaveBurst enhanced by machine learning. Physical Review D, 107(6), 062002.
- Relton P., Virtuoso A., **Bini S.**, Raymond V., Harry I., Drago M., ... Tiwari S. (2022). Addressing the challenges of detecting time-overlapping compact binary coalescences. Physical Review D, 106(10), 104045.
- Vedovato, G., Milotti, E., Prodi, G. A., **Bini, S.**, Drago, M., ... Klimenko, S. (2022). Minimally-modeled search of higher multipole gravitational-wave radiation in compact binary coalescences. Classical and Quantum Gravity, 39, 045001

LIGO-Virgo-KAGRA Collaboration publications

Major contributions

- LIGO, Virgo and KAGRA Collaboration, (2025). GW231123: a Binary Black Hole Merger with Total Mass 190-265 $M_{\odot}$. ApJL 993 L25. Contribution: paper manager, analyst
- LIGO, Virgo and KAGRA Collaboration, (2025). All-sky search for short gravitational-wave bursts in the first part of the fourth LIGO-Virgo-KAGRA observing run. Phys. Rev. D 112, 102005. Contribution: part of the paper writing team and analyst.
- LIGO, Virgo and KAGRA Collaboration, (2024). Search for gravitational-wave transients associated with magnetar bursts in Advanced LIGO and Advanced Virgo data from the third observing run", The Astrophysical Journal 966.137. Contribution: part of the paper writing team and analyst.

Significant contributions

- LIGO, Virgo and KAGRA Collaboration, (2026). GWTC-5.0: Observations from the Second Part of the Fourth LIGO-Virgo-KAGRA Observing Run and Updates to the Gravitational-Wave Transient Catalog, arXiv preprint 2605.27225. Contributions: analyst (mitigation of transient noise overlapping with astrophysical signals and waveform consistency test).
- LIGO, Virgo and KAGRA Collaboration, (2025). GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run, The Astrophysical Journal Letters 1004.2, L22. Contribution: analyst (waveform consistency test).

A complete list of publications is available via ORCID (`https://orcid.org/0000-0002-0267-3562`).

A list of contributions to the LVK Collaboration is available at: `https://laac.docs.ligo.org/recognition/2025/11/21/sophie-bini`

Refereed peer-review research journals

- Physical Review D • The Astrophysical Journal • Nature Astronomy

Spoken languages

• Italian (native speaker) • French (near-native speaker) • English (fluent)